

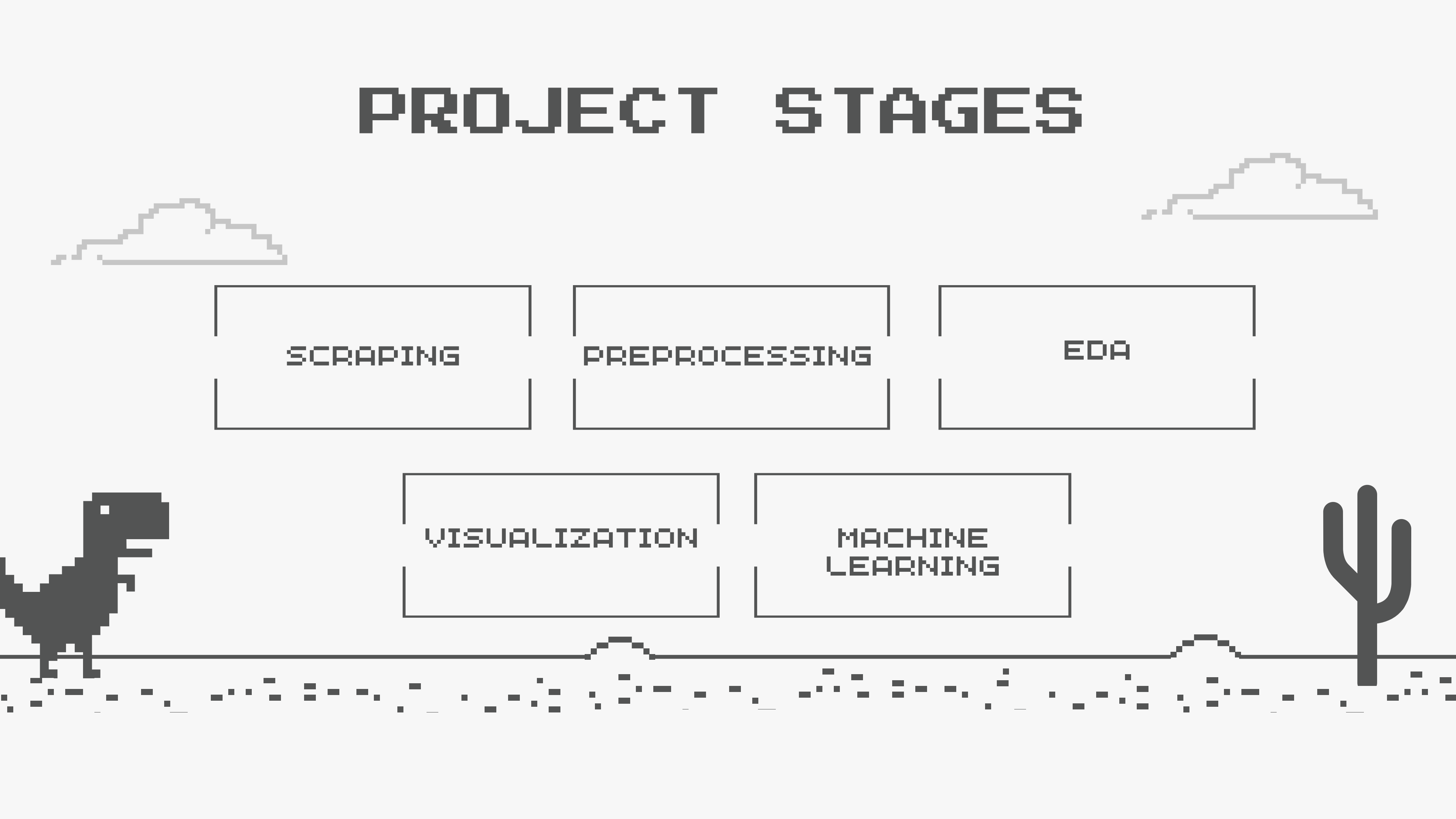
17

UNIVERSITY RANKINGS ANALYSIS

START



PROJECT STAGES

The background features a pixelated landscape. At the top, the title 'PROJECT STAGES' is centered. Below it, five rectangular boxes are arranged in two rows. The top row contains 'SCRAPING', 'PREPROCESSING', and 'EDA'. The bottom row contains 'VISUALIZATION' and 'MACHINE LEARNING'. The bottom of the image shows a horizon line with a pixelated ground, a dinosaur on the left, and a cactus on the right. There are also pixelated clouds in the sky.

SCRAPING

PREPROCESSING

EDA

VISUALIZATION

MACHINE
LEARNING

DATA SCRAPING

WEBSITE



SHANGHAI
RANKING

TOOLS



BEAUTIFUL SOUP



REQUESTS



SELENIUM

CHALLENGES



UNCCLICKABLE BUTTONS



SRAPED DATA

1 TABLE

- 7000+ ROWS
- 9 COLUMNS
- LOW AMOUNT OF MISSING VALUES

2 TABLE

- 3000+ ROWS
- 17 COLUMNS
- A PLETHORA OF MISSING VALUES

WE SRAPED 2 DIFFERENT
TABLES FOR CROSS
ANALYSIS



PREPROCESSED DATA

1 TABLE

- 6000+ ROWS
- 17 COLUMNS
- CORRECTED DATA TYPES

2 TABLE

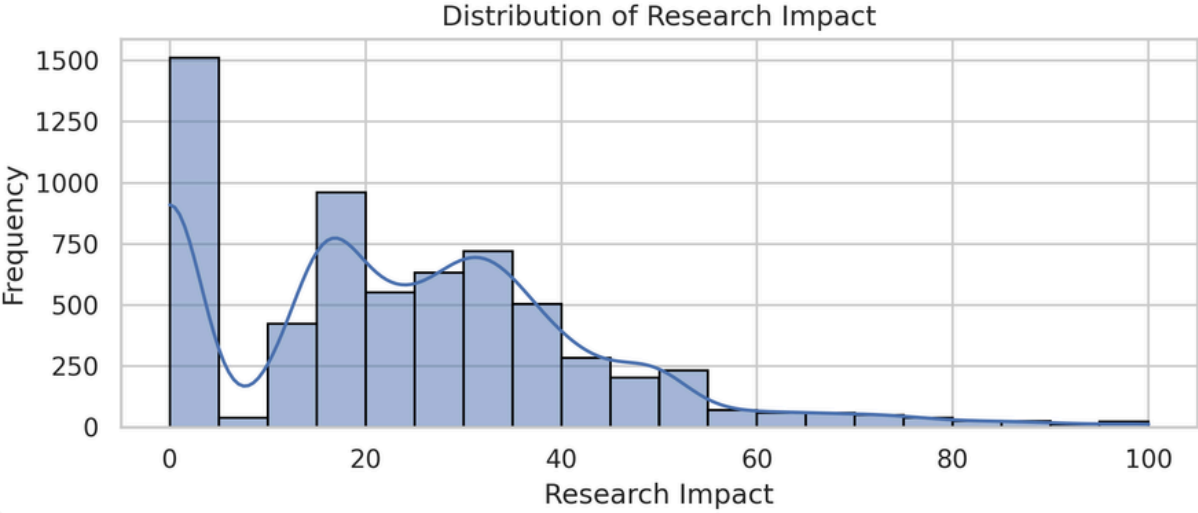
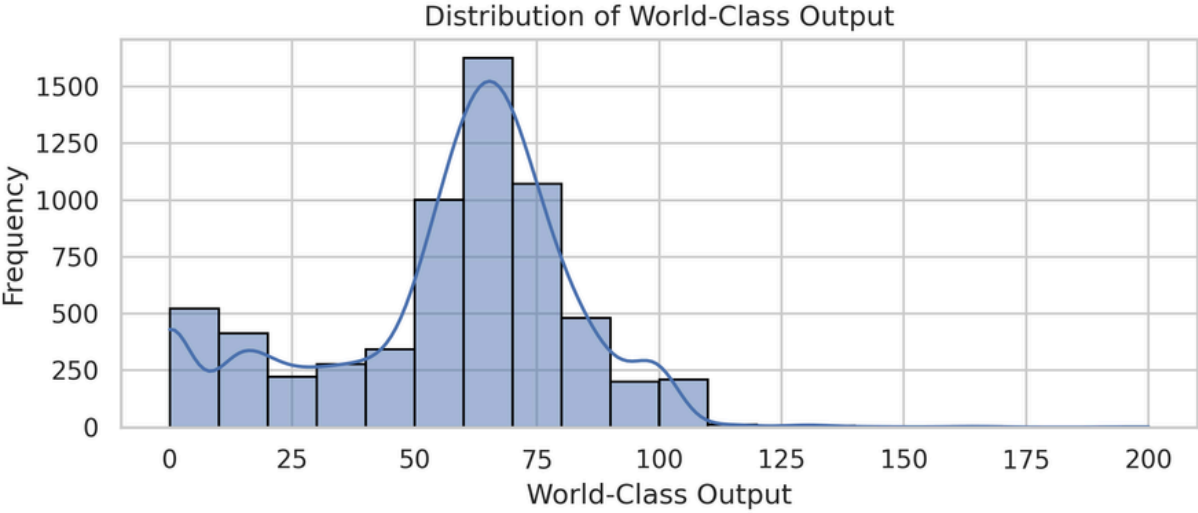
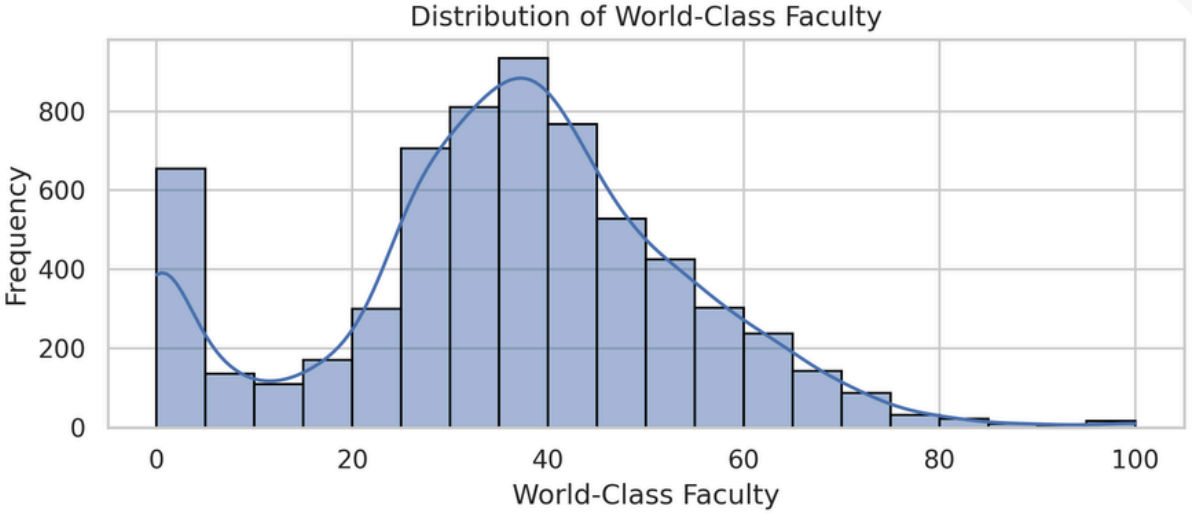
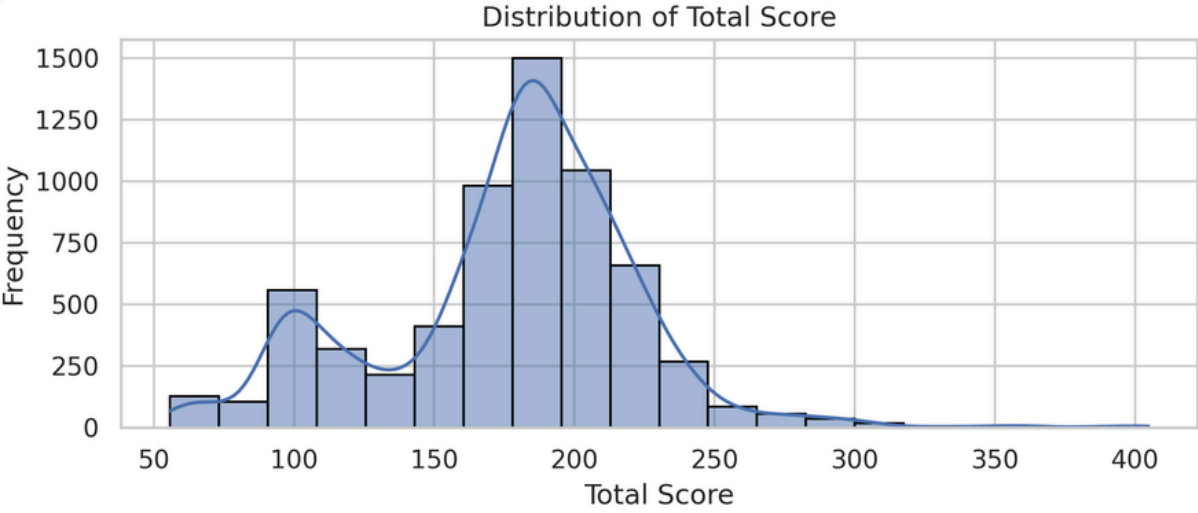
- 3000+ ROWS
- 10 COLUMNS
- MAIN FOCUS —
FOUND YEAR

WE SRAPED 2
DIFFERENT TABLES
FOR CROSS
ANALYSIS

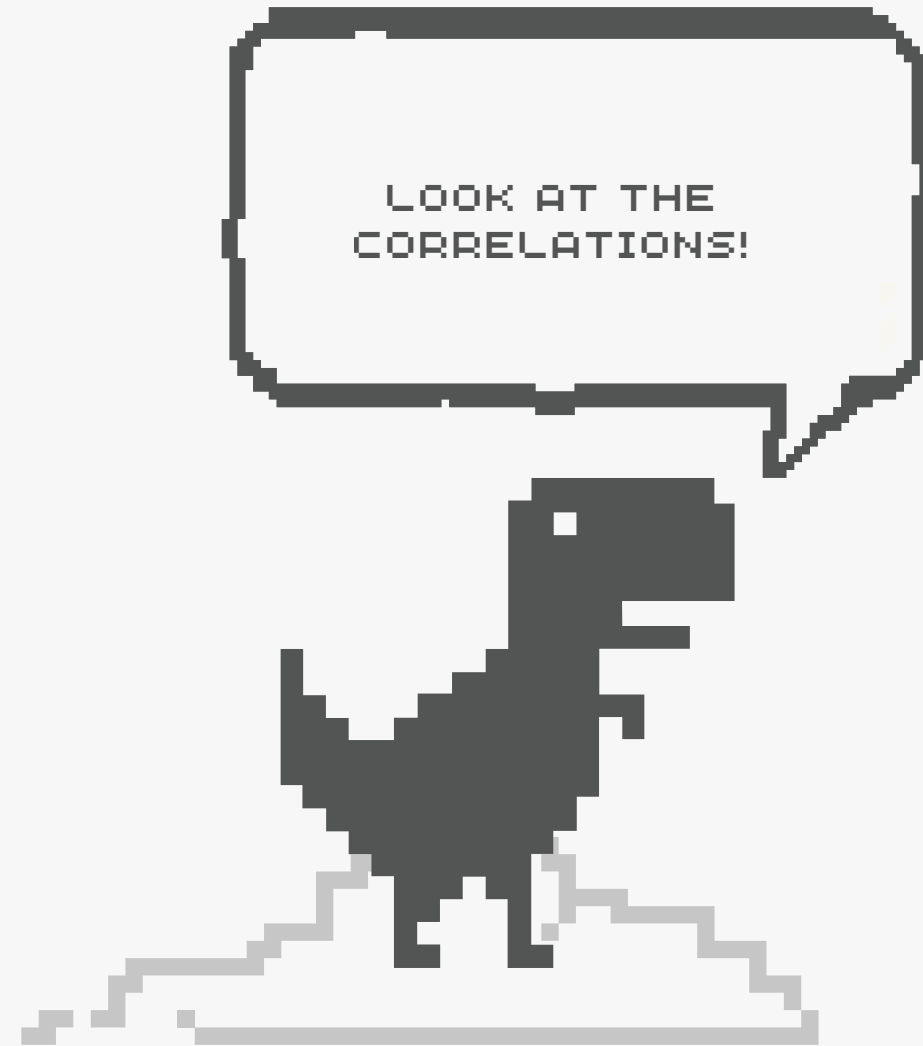


EDA

DISTRIBUTION ANALYSIS



EOA HEATMAP



EDA

INTERSETING HYPOTHESIS

- NULL HYPOTHESIS: ALL CONTINENTS HAVE THE SAME MEAN TOTAL SCORE.
- ALTERNATIVE HYPOTHESIS: EUROPE HAS A HIGHER MEAN TOTAL SCORE.
- METHODOLOGY: ONE-WAY ANOVA FOLLOWED BY TUKEY'S HONEST SIGNIFICANT DIFFERENCE (HSD) POST-HOC TEST.



STATISTICAL RESULTS



F-STATISTIC: 39.6531



P-VALUE: 0.0000....

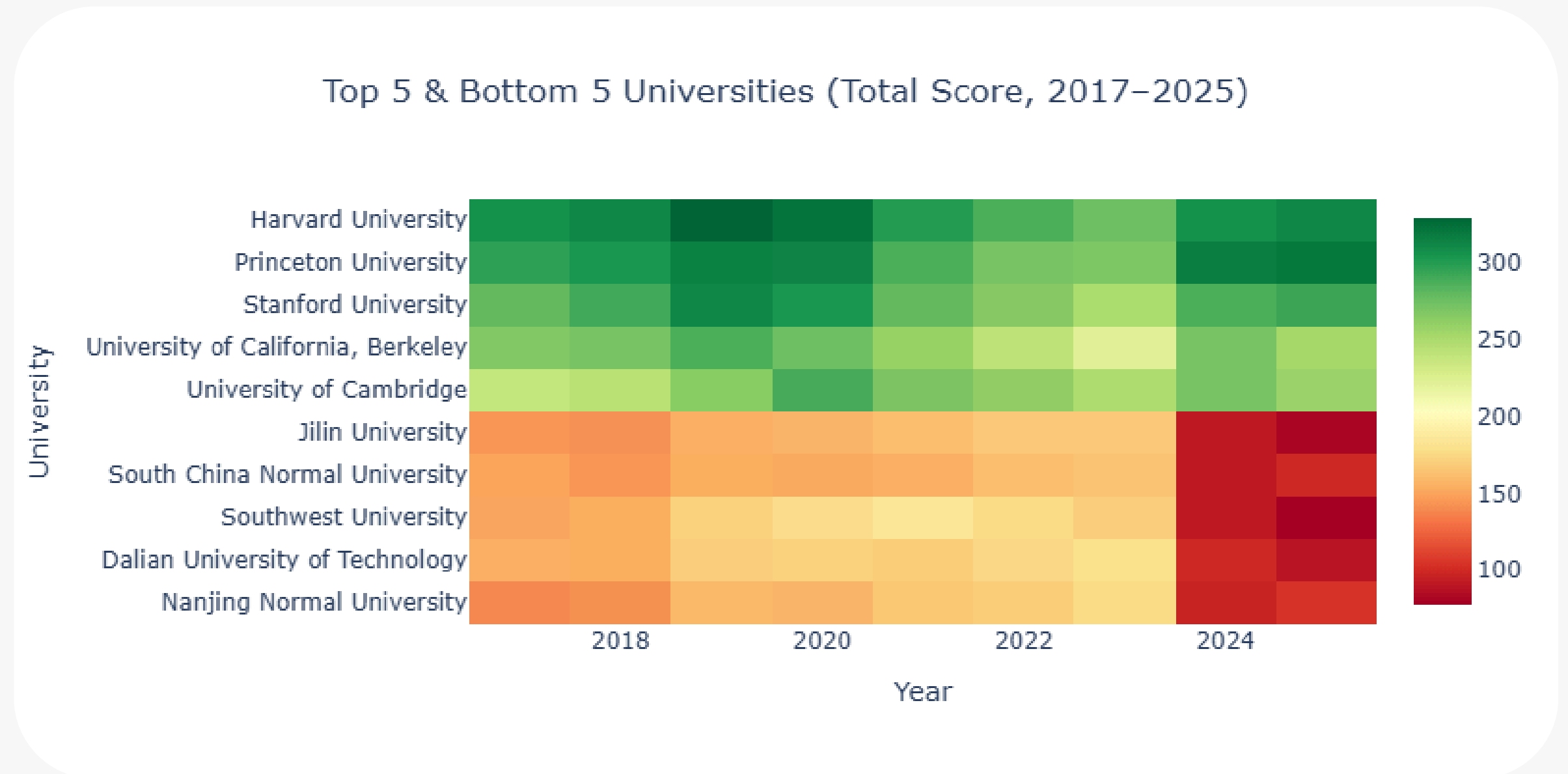
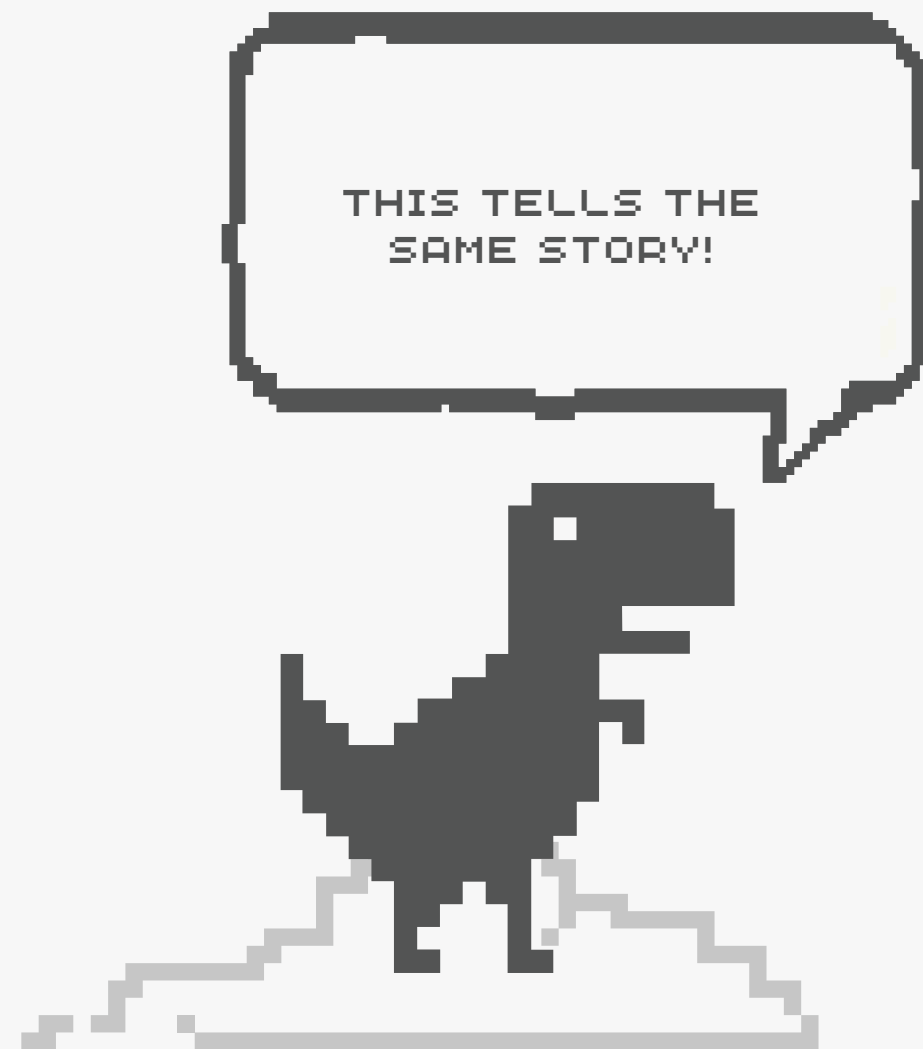


REJECT!!

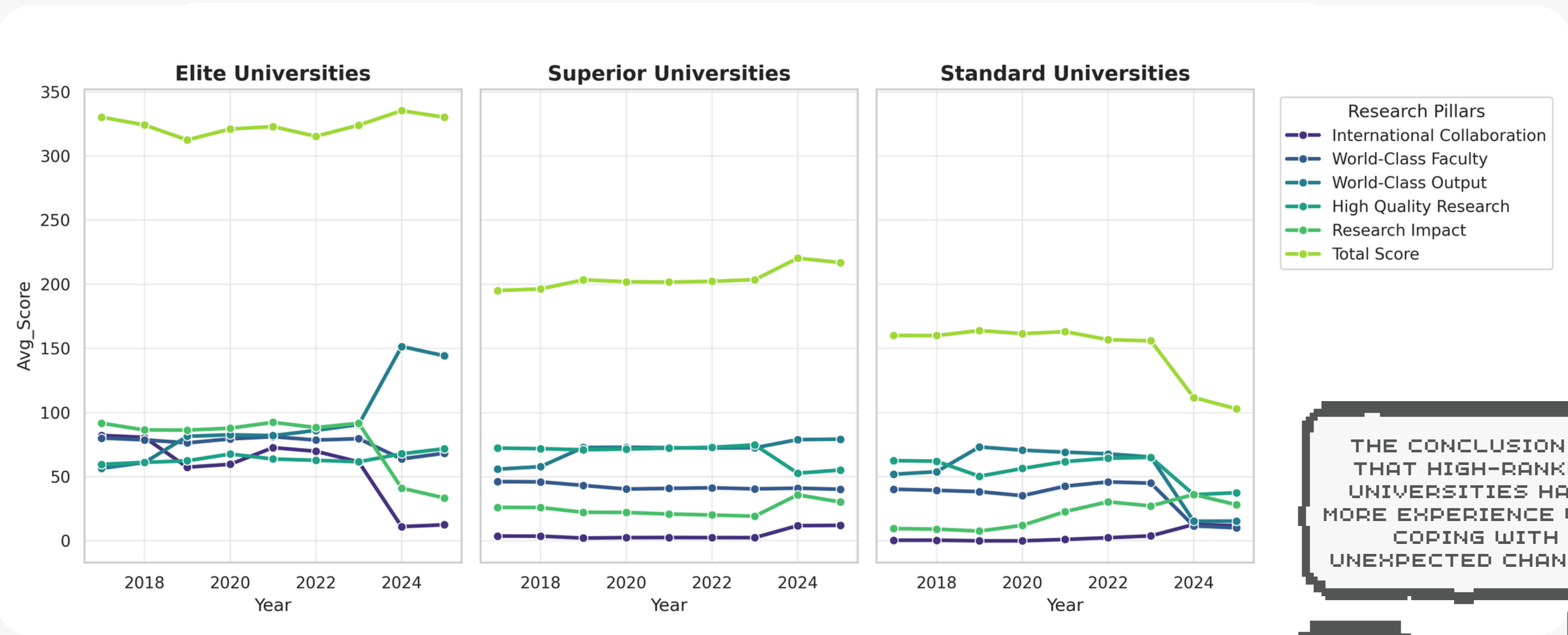
- THIS CONFIRMED THAT AT LEAST ONE CONTINENT PERFORMS DIFFERENTLY.
- TUKEY HSD INSIGHT: WHILE EUROPE PERFORMED STRONGLY, THE PAIRWISE COMPARISON REVEALED THAT NORTH AMERICA ACTUALLY HELD THE HIGHEST MEAN DIFFERENCE IN TOTAL SCORES.

EDA

TOP & BOTTOM 5

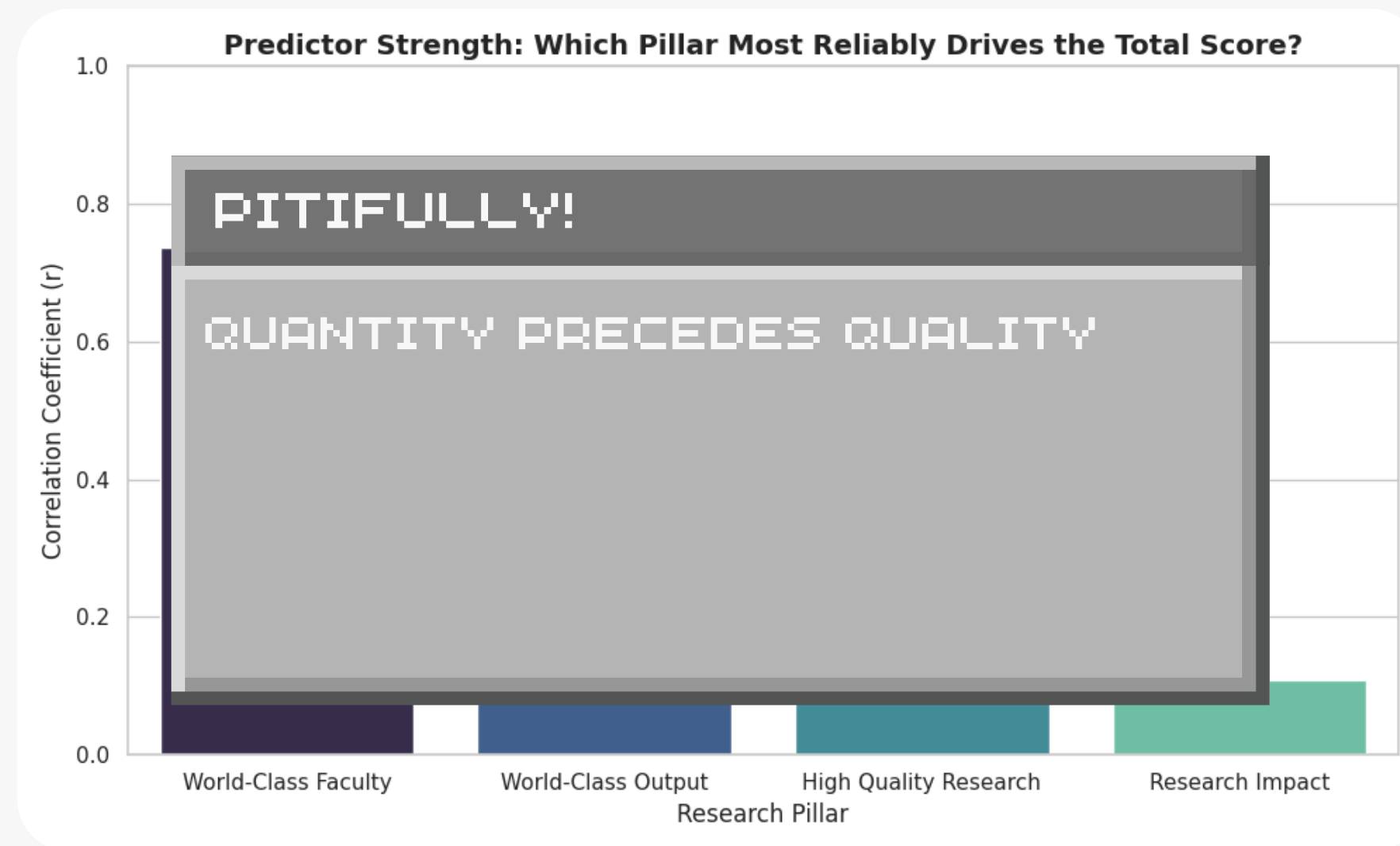


GLOBAL DISTRIBUTION OF UNIVERSITY SCORES OVER TIME



THE «QUALITY VS QUANTITY» DILEMMA

WHICH IS BETTER:
PUBLISHING MANY
PAPERS IN TOP
JOURNALS, OR
PUBLISHING FEWER BUT
HIGH QUALITY?

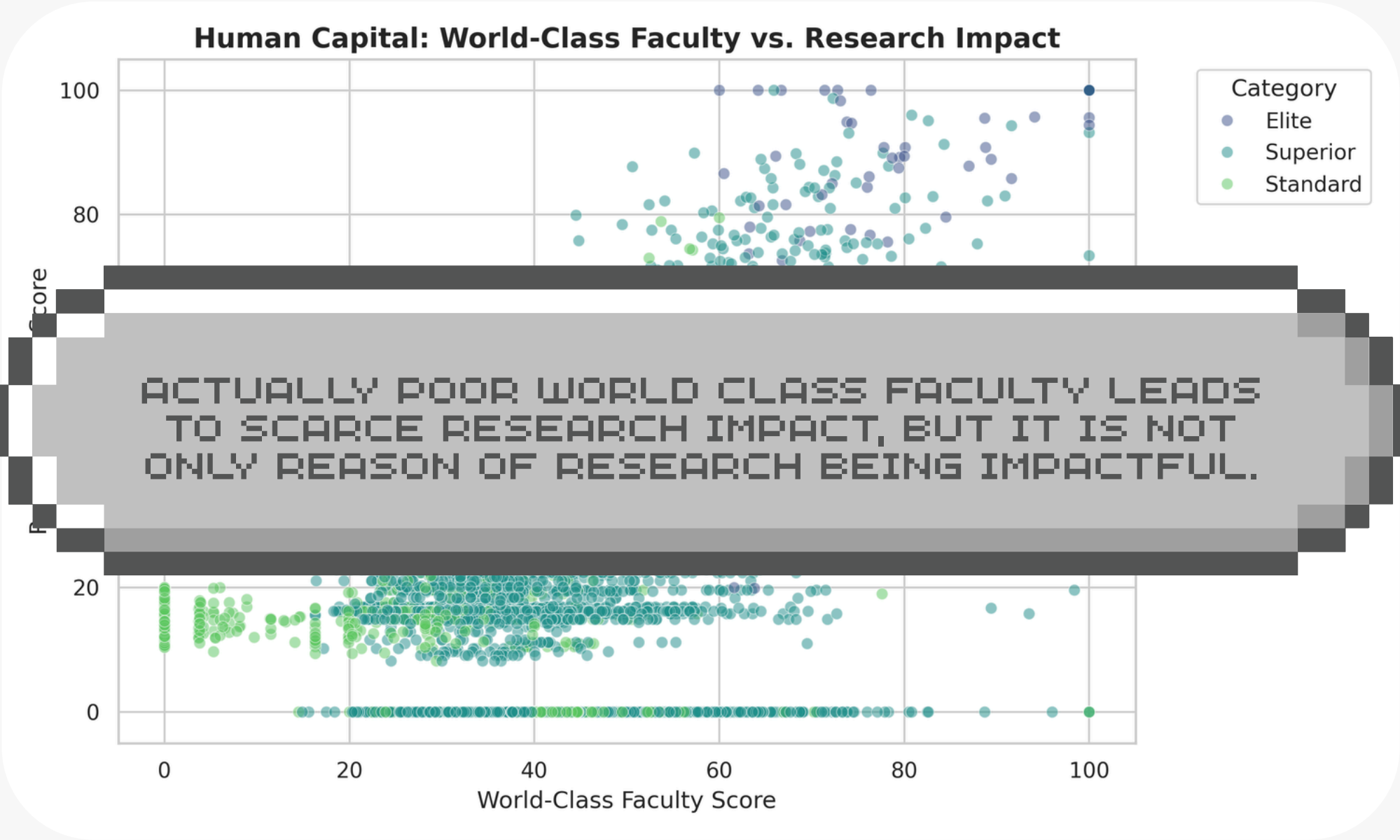


WHY GUESS IF
TEST?



THE «HUMAN CAPITAL» HYPOTHESIS

DOES THE PRESENCE OF «WORLD-CLASS FACULTY» ACT AS A «MULTIPLIER» FOR RESEARCH IMPACT, OR IS IT JUST A STATUS SYMBOL?

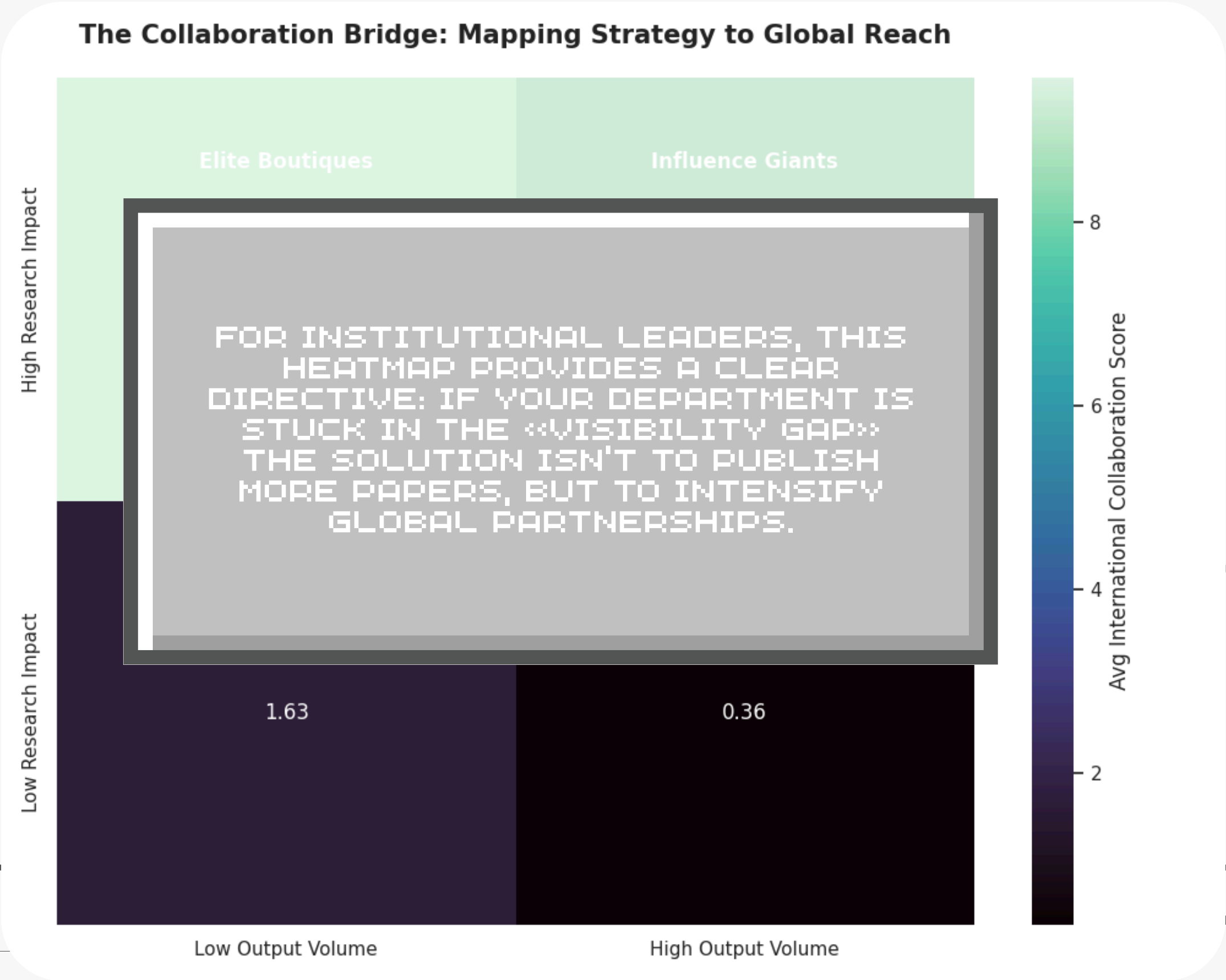


YOU KNOW MY POINT...



WORLD-CLASS OUTPUT VS RESEARCH IMPACT

CAN A UNIVERSITY HAVE
HIGH WORLD-CLASS
OUTPUT BUT LOW
RESEARCH IMPACT, AND
VICE-VERSA?

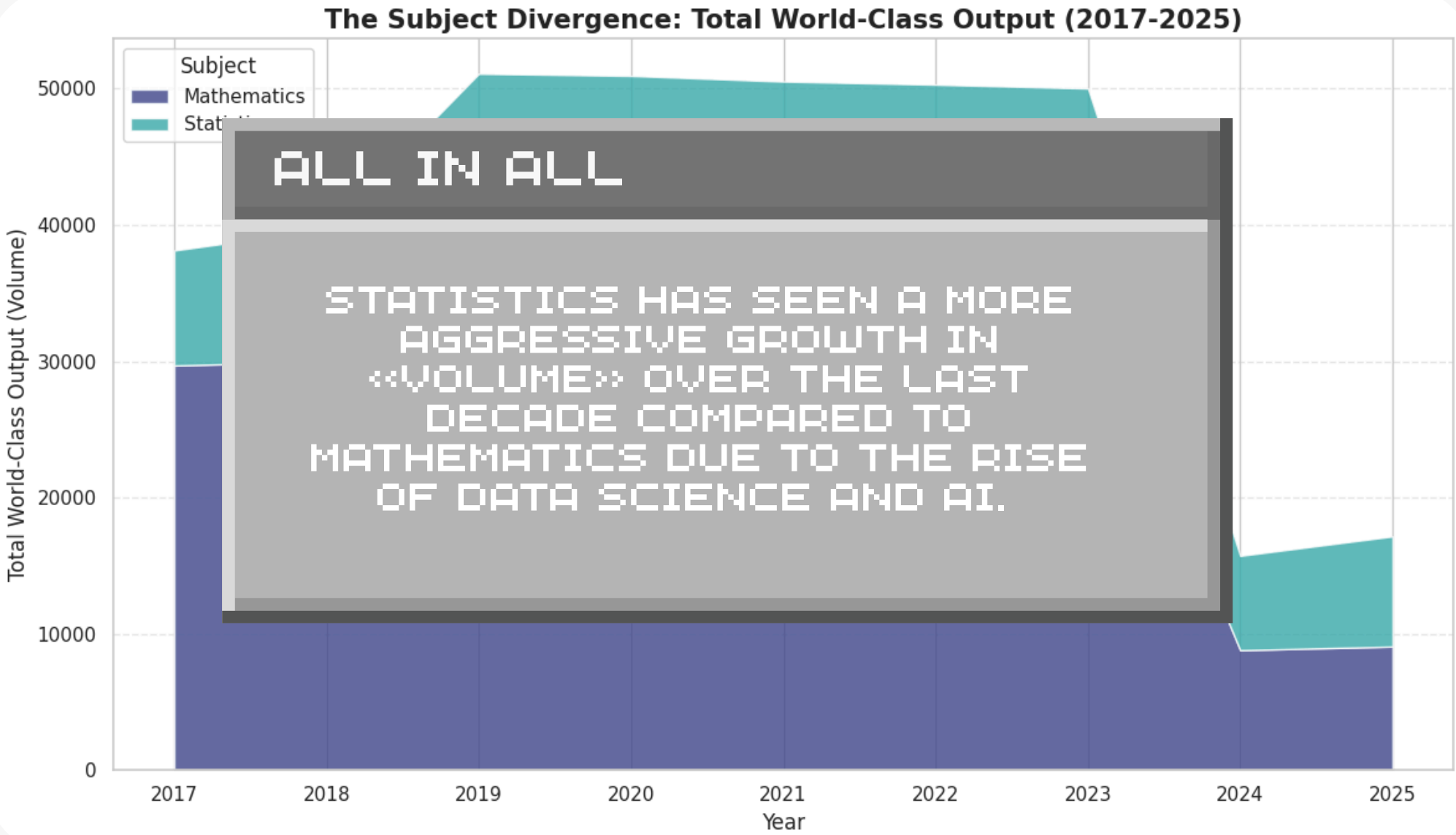
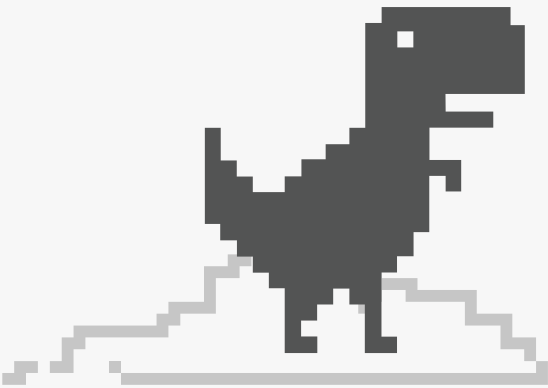


WHY NOT?



THE SUBJECT DIVERGENCE

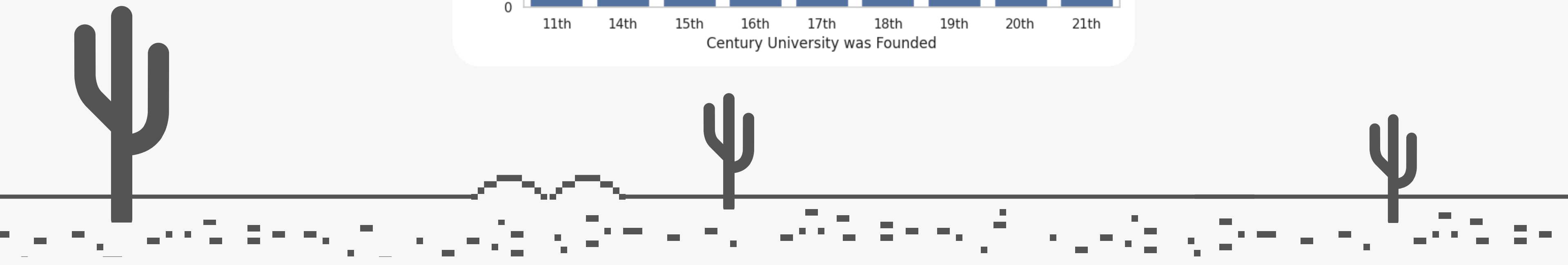
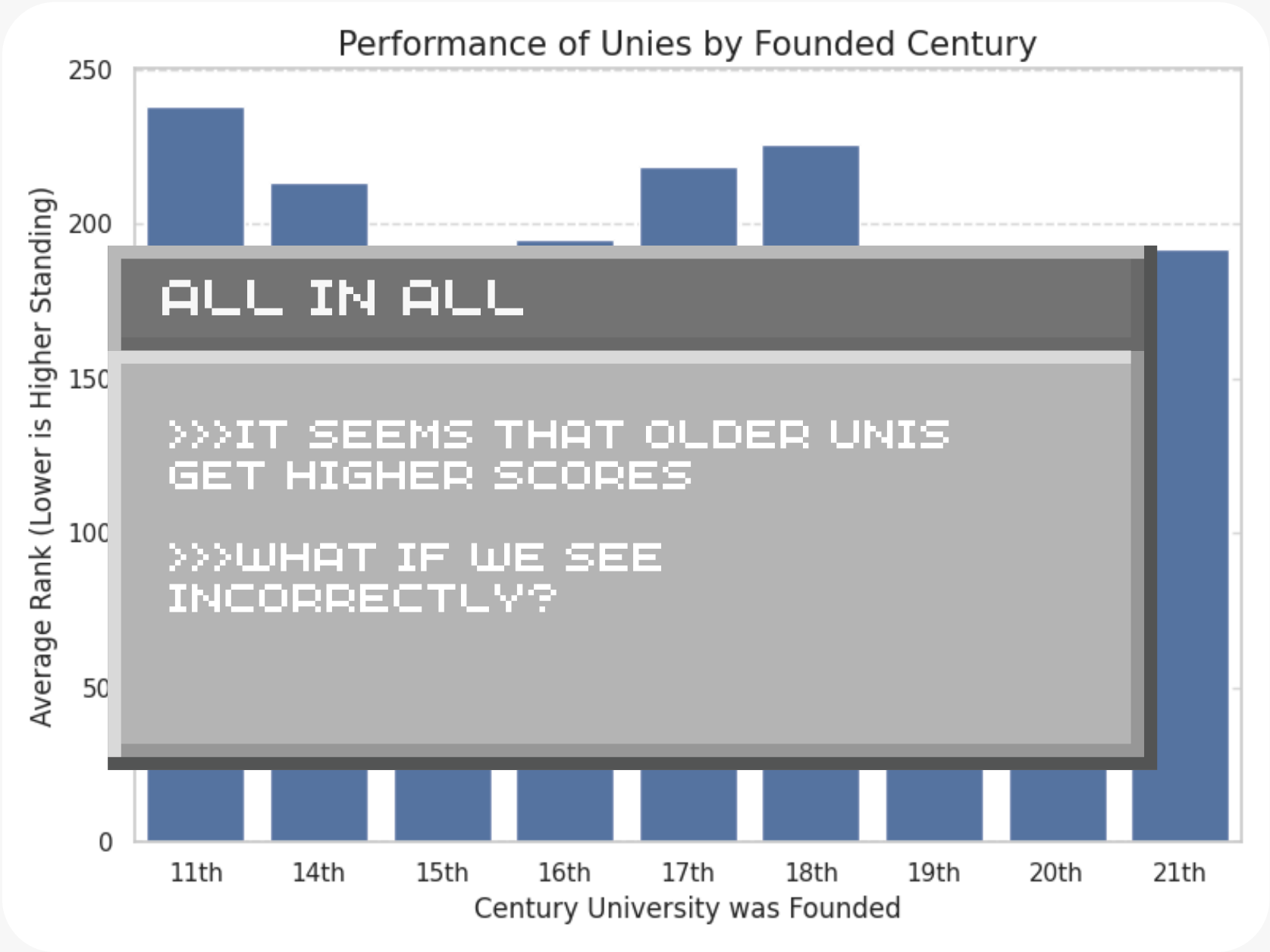
STATISTICS OR
MATHEMATICS?



WHAT KIND OF
QUESTION IS IT?



PERFORMANCE OF UNIES BY FOUNDED CENTURY



PROVE INTERESTING HYPOTHESIS

- NULL HYPOTHESIS: TOTAL SCORE MEANS OF OLD AND NEW UNIS ARE EQUAL.
- ALTERNATIVE HYPOTHESIS: OLD UNIS HAS GREATER TOTAL SCORE MEAN.
- METHODOLOGY: WELCH'S T-TEST (INDEPENDENT T-TEST WITH UNEQUAL VARIANCE) AFTER MERGING THE RANKING AND FOUNDING YEAR DATASETS.



STATISTICAL RESULTS

🖱️ T-STATISTIC: 3.369

🖱️ P-VALUE < 0.05

🖱️ REJECT!!

- THE STATISTICAL EVIDENCE STRONGLY SUPPORTS THE ALTERNATIVE HYPOTHESIS THAT OLDER UNIVERSITIES (PRE-1950) MAINTAIN SIGNIFICANTLY HIGHER TOTAL SCORES THAN NEWER INSTITUTIONS.
- THIS SUGGESTS THAT FACTORS LIKE LONG-TERM REPUTATION, ESTABLISHED RESEARCH INFRASTRUCTURE, AND ALUMNI NETWORKS CONTRIBUTE TO SUSTAINED HIGH RANKINGS.

LINK TO
ANIMATION
PLOT



ABOUT ML MODEL



PROJECT OBJECTIVE:
COMPARATIVE ANALYSIS
OF ACADEMIC
DISCIPLINES USING
MACHINE LEARNING



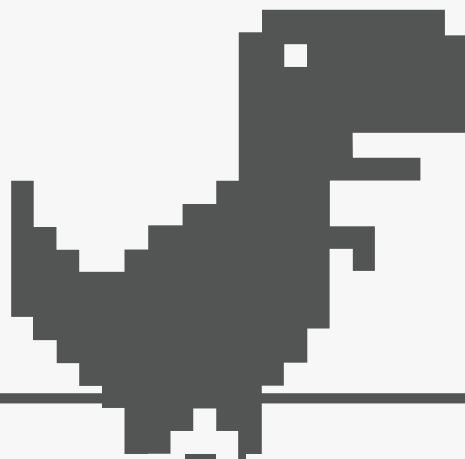
GOAL:
TO DETERMINE WHETHER
«MATHEMATICS» AND «STATISTICS» CAN
BE DISTINGUISHED BASED ON THEIR
ACADEMIC PERFORMANCE METRICS



MODEL:
RANDOM FOREST
CLASSIFIER



EVALUATION:
CONFUSION MATRIX



ABOUT ML MODEL



HIGH MODEL
ACCURACY

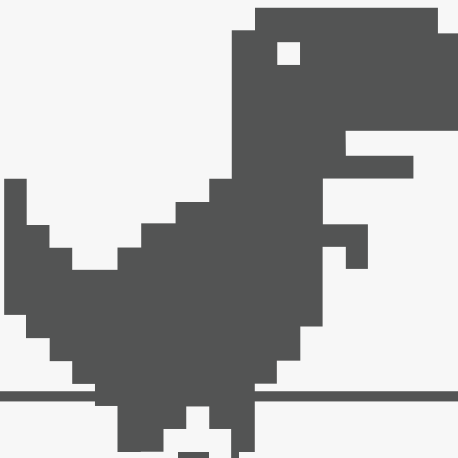
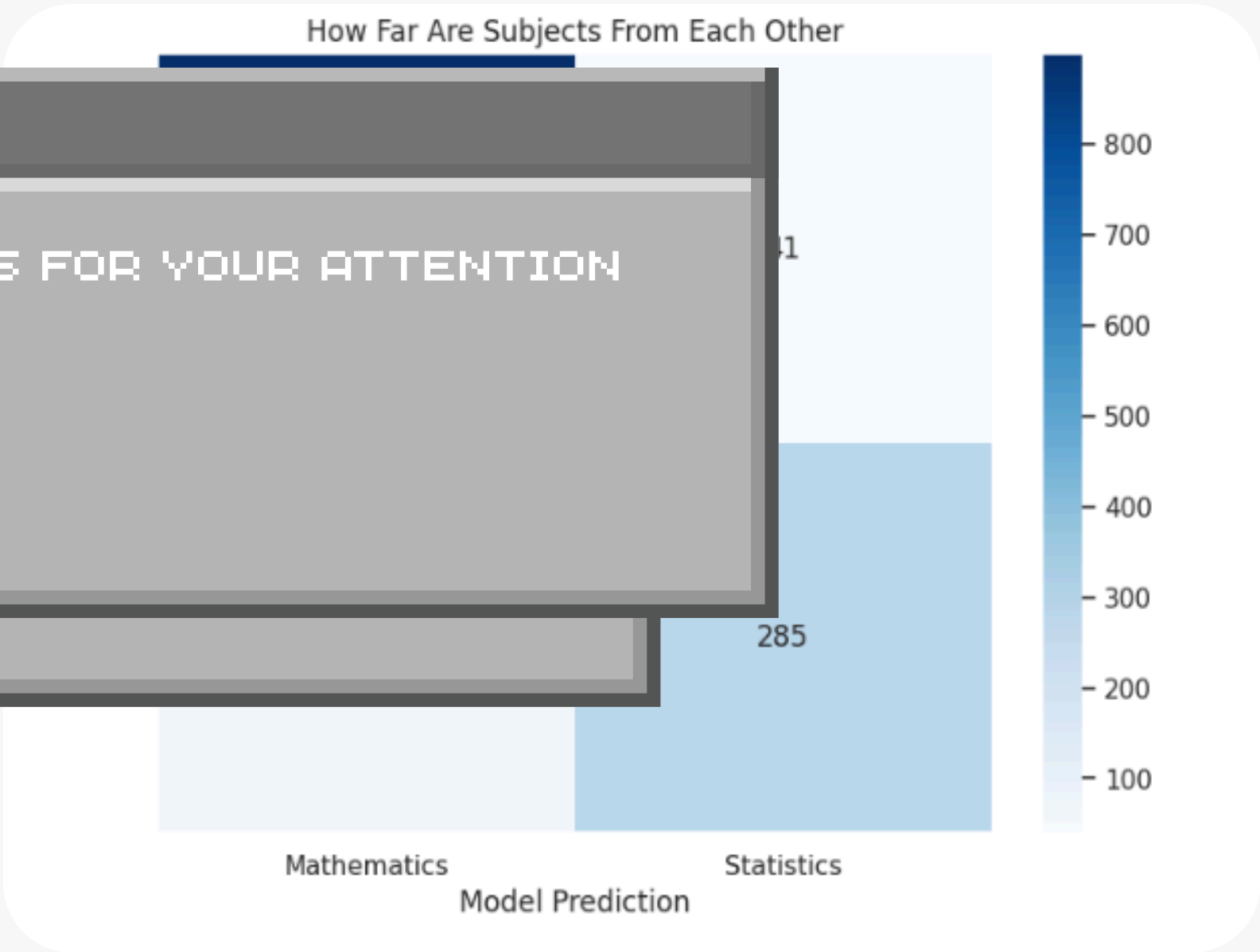


KEY DIFFERENTIATING
RESEARCH IMPACT

BYE

>>> THANKS FOR YOUR ATTENTION

>>> SEVERAL MAJOR PAPER ENROLLMENT RESEARCH DRIVERS.



GAME OVER

